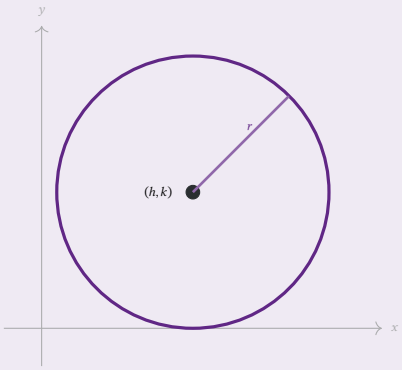
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 21

RULE · Equation of a circle (aylana tenglamasi)

$$(x-h)^2+(y-k)^2=r^2$$

Center (h,k), radius r.

$(x+6)^2$ means $h=-6$ – the sign in the equation is opposite the center coordinate.



RULE · Arc length (yoy uzunligi)

$$\text{arc length}=\frac{\text{central angle}}{360^\circ}\times 2\pi r$$

$r=9,\ \text{angle}=60^\circ:\quad \frac{60}{360}\times 18\pi=3\pi$

Arc length is a distance – without the radius, only the angle (arc measure) can be given, not a length.

RULES SHEET – 2/2

RULE · Sector area (sektor yuzi)

$$\text{sector area} = \frac{\text{central angle}}{360^\circ} \times \pi r^2$$

$r = 6$, angle $= 120^\circ$: $\frac{120}{360} \times 36\pi = 12\pi$

Sector area uses πr^2 (area); arc length uses $2\pi r$ (distance) – don't swap them.

RULE · Circle equation by completing the square

General form $x^2 + y^2 + Dx + Ey + F = 0$ – complete the square on x and y separately.

$x^2 + y^2 - 6x + 4y - 12 = 0$

$(x^2 - 6x) + (y^2 + 4y) = 12 \implies (x-3)^2 + (y+2)^2 = 12 + 9 + 4$

$(x-3)^2 + (y+2)^2 = 25 \implies$ center $(3, -2)$, radius 5

$(y+2)^2 = (y-(-2))^2$, so $k = -2$ – same sign-flip trap as the center form.